

Jenkins Docker Homework Assignments

This document contains five hands-on Jenkins homework assignments. They progress from running jobs on the local master to using ephemeral Docker Cloud agents. Each assignment specifies setup steps, tasks, and evidence requirements.

HW1 — 'Hello, Master': Freestyle job on local master

Goal: Prove Jenkins is up and can run builds on the master (controller) container.

Tasks:

- Create a Freestyle job `master-hello`.
- Restrict it to run on master node.
- Add a shell step: `echo 'Hello from master on $(hostname)', 'whoami', 'df -h'`.
- Build it.

Evidence to submit:

- Screenshot of Build #1 Console Output.
- Screenshot of Build Executor Status showing master.
- Screenshot or config.xml of job configuration.

HW2 — Ephemeral Docker agent via Docker plugin

Goal: Use Docker Cloud to spin up an on-demand build agent container.

Tasks:

- Configure Docker Cloud with `tcp://172.22.0.2:2375` (The IP address may be different in your case. Please use your container IP address).
- Add agent template with label `docker-agent`.
- Create a job `docker-agent-echo` with simple echo shell.
- Restrict this job to be run only with docker-agent label
- Create another job `docker-agent-github` with the source code from <https://github.com/qingsong-cstu/jenkinsDemo> and to run a shell command `uname -a`.
- Restrict this job to be run only with docker-agent label as well.

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Evidence to submit:

- Screenshot of Nodes showing ephemeral agent online.
- Console log snippet with hostname and `java -version`.
- Screenshot of agent template.

HW3 (Bonus) — Create Jenkins agent with Python enabled.

Goal: Use Docker Cloud with a custom toolchain image.

Tasks:

- Based on Jenkins image: jenkins/agent:latest-jdk21, add Python3 capability into this image. (Customized Dockerfile)
- Build this new agent docker image with Python capability
- Add agent template with label `docker-agent-python`.
- Create a job to run “python3 helloworld.py” from GitHub: <https://github.com/qingsong-cstu/JenkinsDemo>.
- Restrict this job to be run only on the agent with label `docker-agent-python`

Evidence to submit:

- Console output showing python versions.
- Screenshot of Nodes showing ephemeral `docker-agent-python` agent.
- Console output showing the output of helloworld.py.